

Complements of Operational Research

2.1 Project planning: CPM

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Why plan a project?

A **project** is a set of activities with a start and an end, linked by precedence relations.

Examples where these methods are applied:

- construction of a building;
- research and development of a product;
- film production;
- shipbuilding, aircraft building;
- installation of an information system;
- advertising or political campaigns.

Questions the manager has to answer:

1. How long will the project take?
2. Which activities **cannot** be delayed?
3. How much can the others be delayed?
4. What if the durations are uncertain? (2.2)
5. How much does it cost to speed up? (2.3)

CPM and PERT

Critical Path Method (DuPont, 1957) and *Program Evaluation and Review Technique* (US Navy, 1958): the same network model, with certain (CPM) or uncertain (PERT) durations.

The example of this chapter: renovating a laboratory

Activity	Predecessors	Dur. (wk.)
A Design and permits	—	3
B Ordering the equipment	A	4
C Removing the old equipment	A	2
D Electrical installation	C	3
E Flooring and painting	C	2
F Installing the equipment	B, D, E	3
G Network and software	D	2
H Testing and training	F, G	2

Precedence: “F can only start after B, D and E have finished”.

Sum of the durations: 21 weeks. But some activities can run **in parallel**.

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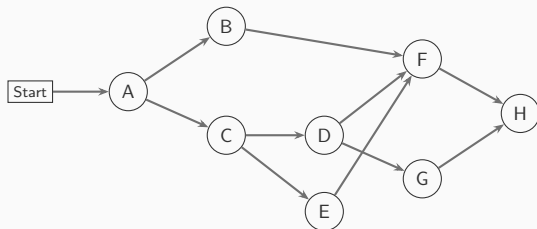
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To answer, the project is represented by a **network**.

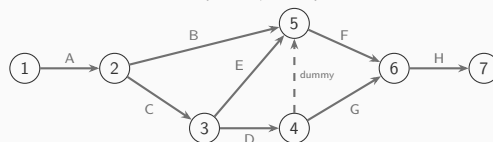
Two representations of the network

AON (*activity on node*): each activity is a node; the arrows are the precedences.



AON: simpler to draw; it is the one we use in the calculations.

AOA (*activity on arrow*): each activity is an arrow; the nodes are events (finish/start).



AOA: the **dummy activity** (duration 0, dashed) is needed because F depends on D, but G depends only on D — without it, G would also depend on B and E.

The four dates of each activity

ES	A	EF
LS	d	LF

E = *early*, L = *late*, S = *start*, F = *finish*;
 d = duration.

Forward pass (earliest dates)

- initial activities: $ES = 0$;
- $ES = \max\{EF \text{ of predecessors}\}$, $EF = ES + d$.

Project duration: $T = \max EF$.

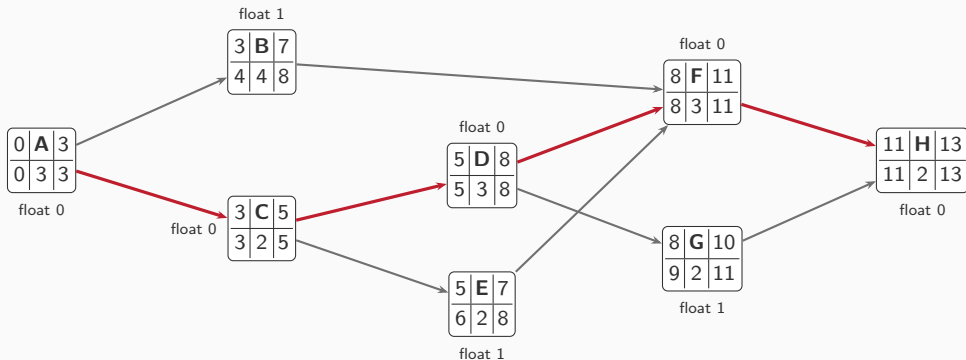
Backward pass (latest dates)

- final activities: $LF = T$;
- $LF = \min\{LS \text{ of successors}\}$, $LS = LF - d$.

Float

float = $LS - ES = LF - EF$: how much the activity can be delayed without delaying the project.

Forward and backward pass



F: $ES = \max\{EF_B, EF_D, EF_E\} = \max\{7, 8, 7\} = 8$.

D: $LF = \min\{LS_F, LS_G\} = \min\{8, 9\} = 8$.

Project duration: 13 weeks (not 21).

Critical path: A, C, D, F, H (float 0, in red).

Critical path = longest path

All paths from start to finish:

Path	Duration
A-B-F-H	$3 + 4 + 3 + 2 = 12$
A-C-D-F-H	$3 + 2 + 3 + 3 + 2 = \mathbf{13}$
A-C-D-G-H	$3 + 2 + 3 + 2 + 2 = 12$
A-C-E-F-H	$3 + 2 + 2 + 3 + 2 = 12$

Two ways of reaching the same result

The project duration is that of the **longest path**; its activities are those with zero float.

Listing paths only works for small networks (the number of paths grows very fast); the forward and backward passes always work.

Beware: here the other three paths have only **1 week** of float. A small delay in B, E or G changes the critical path.

What to do with the floats?

Critical activities (A, C, D, F, H)

Any delay delays the project. These are the ones the manager follows closely and where the best resources go.

Activities with float (B, E, G)

They can start anywhere between *ES* and *LS* without delaying the project. That margin allows **resource levelling** (not having all teams busy at the same time).

Careful: float is **shared** along a path. If B uses its week of float (starting in week 4), the path A–B–F–H becomes critical.

Question: ordering the equipment (B) is delayed by 2 weeks. How much is the project delayed?

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B now takes 6: $EF_B = 9 > LS_F = 8$. The project becomes **14 weeks** (1 week of delay: B's float absorbed the other one).

Example 1

Activity	Predecessors	Duration (wk.)
A	—	2
B	—	3
C	A	2
D	A, B	4
E	C	4
F	C	3
G	D, E	5
H	F, G	2

- Build the AON and AOA networks.
- Compute ES , EF , LS , LF and the float of each activity.
- Give the critical path and the project duration.
- List all paths and confirm that the critical one is the longest.

To solve in class (2/2)

Example 2 — construction of a building (Hillier & Lieberman)

Activity		Pred.	Dur. (wk.)	Activity		Pred.	Dur. (wk.)
A	Excavation	—	2	H	Exterior painting	E, G	9
B	Foundations	A	4	I	Electrical work	C	7
C	Walls	B	10	J	Partitions	F, I	8
D	Roof	C	6	K	Flooring	J	4
E	Exterior plumbing	C	4	L	Interior painting	J	5
F	Interior plumbing	E	5	M	Exterior finishing	H	2
G	Boundary walls	D	7	N	Interior finishing	K, L	6

The company pays a penalty of 300 000 \$ if the work finishes after 47 weeks and receives a bonus of 150 000 \$ if it finishes before 40.

- Determine the project duration and the critical path.
- Does the company risk the penalty? Can it earn the bonus? Which activities would you watch closely?

KEY TAKEAWAYS — 2.1

1. A project is a network of activities with **precedences**; the sum of the durations is not the project duration.
2. **AON** networks (activities on the nodes) and **AOA** networks (on the arrows, with dummy activities when needed).
3. **Forward**: $ES = \max EF$ of the predecessors. **Backward**: $LF = \min LS$ of the successors.
4. $\text{float} = LS - ES = LF - EF$; **critical path** = zero float = longest path.
5. Floats give room to manage resources, but they are shared along the paths.

What if the durations are not known with certainty? → 2.2 PERT.

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Project Management with PERT/CPM.
- Heizer, J., Render, B., *Operations Management*, Pearson — ch. 3, Project Management.
- PMI, *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*, 5th ed., 2013.

Resources

- Exercise sheet 2 — Project planning. Practical assignment no. 1 (PERT and CPM with *crashing*).